JAN–C–716

25 JANUARY 1949

NATIONAL MILITARY ESTABLISHMENT SPECIFICATION

CARTRIDGE, BALL, LONG RIFLE, CALIBER .22, M24

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments.

A. APPLICABLE SPECIFICATIONS AND DRAWINGS

A–1. The following drawings, including all drawings and specifications listed thereon, and specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

NATIONAL MILITARY ESTABLISHMENT SPECIFICATION

JAN–A–625—Ammunition, Small Arms, General Specification for.

U. S. ARMY SPECIFICATION

100–2—Standard Specification for Marking Shipments by Contractors.

BUREAU OF SUPPLIES AND ACCOUNTS PUBLICATION

Navy Shipment Marking Handbook [1]

U. S. ARMY ORDNANCE DEPARTMENT DRAWINGS

B7640995—Cartridge, Ball, Caliber .22, Long Rifle, M24.
42–23–2—Rifle, U. S., Caliber .22, M2.
49–15–1—Gage, Pressure, Caliber .22, M52.
A47753—Cylinders, Pressure.
A7575482—Cups, Obturating.
A7575484—Pistons.
C7670116—Gages, Pressure, Piston Hole Assembly.
B7634883A—Gages, Pressure, Piston Seat.
B7634883B—Gages, Pressure, Piston Seat.
A7575483—Gages, Pressure, Piston.
C7669514—Tools, Seating and Knockout.
C7675767—Packing Box, Caliber .22, Ammunition (Markings).
FA–26631—Machine, Bullet Pull, Testing, Case Stop Assembly.
FA–26632—Machine, Bullet Pull, Testing Jaws.
FA–26629—Machine, Bullet Pull, Testing, Case Holder.
FA–26630—Machine, Bullet Pull, Testing, Case Holder Base.

COMMERCIAL DRAWINGS

Rifle, Winchester, Caliber .22, M52.
Rifle, Mossberg, Caliber .22, M44.

[1] Applicable only to Navy shipments.

Source: https://assist.dla.mil -- Downloaded
Check the source to verify that this is the cur

Rifle, Remington, Caliber .22, Model 37.
Shotgun, Rifle, Stevens, .410-Gage, Caliber .22, Over and Under Barrel.
Rest, Approved.

B. TYPE AND GRADE

B–1. This specification covers cartridge, ball, long rifle, caliber .22, M24, in one grade, as follows:

Jacketed bullet—regular velocity.

C. MATERIAL AND WORKMANSHIP

C–1. Material and workmanship shall be in accordance with best commercial practice for this type of ammunition.

C–2. All cartridges shall be uniformly assembled in workmanlike manner, free of processing and handling defects, except as provided herein.

D. GENERAL REQUIREMENTS

D–1. See section E.

E. DETAIL REQUIREMENTS

E–1. *Cartridge components.*—

E–1a. *General.*—The requirements of National Military Establishment Specification JAN–A–625, section D, shall apply with the exception of paragraphs D–3b, D–4, D–5a, D–5b, and D–5d.

E–1b. *Cartridge case.*—The metal of the cartridge case shall be brass.

E–1c. *Propellent powder.*—The powder shall be a smokeless type of approved composition.

E–1d. *Priming.*—The priming composition shall be nonmercuric and noncorrosive and suitable for the type of cartridge used.

E–1e. *Bullet.*—The bullet shall consist of a jacketed type, with physical characteristics in accordance with the applicable drawing.

E–1f. *Visual defects.*—

E–1f (1). The acceptable quality level of the ammunition, with respect to visual defects, shall be as follows:

Major—0.10 percent
Minor—0.50 percent

E–1f (2). The standard for visual defects listed in table I shall be that furnished or approved by the contracting officer. (See par. E–3.)

[JAN–C–71C]

Source: https://assist.dla.mil -- Downloaded:
Check the source to verify that this is the curre

TABLE I.—*Visual defect classification.*

Defect	Major	Minor	Major or minor
Cartridge:			
Corroded	X		
Crooked [1]			X
Case:			
Dent			X
Mashed [1]			X
Split	X		
Perforated	X		
Draw scratch			X
Draw rough	X		
Scratch		X	
Scaly metal			X
Fold		X	
Wrinkle		X	
Bulge		X	
Defective head [2]			X
Defective mouth		X	
Bullet:			
Dent		X	
Scratch		X	
Gash	X		
Upset point		X	
Flat point		X	
Split bullet jacket	X		
Scaly metal [1]			X
Exposed lead [1]			X
Loose bullet		X	

[1] Classified as major if the cartridges fail to chamber in an M2 rifle.
[2] Classified as major if defect is such as to cause misfire.

E–2. *Bullet pull.*—The force required to pull any one bullet from its case shall not be less than 20 pounds. The average force required to pull bullets from the cartridges shall not be less than 35 pounds.

E–3. *Pressure.*—The average pressure shall not exceed 24,000 pounds per square inch.

E–4. *Velocity.*—The basic velocity at 25 feet from the muzzle shall be 1125 plus or minus 25 feet per second.

E–5. *Accuracy.*—The extreme spread obtained in targets taken at a distance of 100 yards and measured center to center shall not exceed 2.20 inches.

E–6. *Noncorrosiveness.*—When 10 cartridges have been fired in a barrel, submitted to the humidity oven and normalized, no rust shall be detected upon visual examination of the bore.

E–7. *Heat.*—

E–7a. *Accuracy.*—Targets of all cartridges fired as part of the heat test shall not give an extreme spread in excess of 0.50 inch over that listed in paragraph E–5.

E–7b. *Velocity.*—The average velocity of the cartridge fired after being subjected to this test shall not be more than 50 feet per second below the instrumental velocity obtained with the cartridges held at normal temperature when fired according to paragraph F–7c.

E–8. *Function and casualty.*—Total defects, resulting from firings in this test and all other tests, shall not exceed a maximum of eight,

Source: https://assist.dla.mil -- Downloaded: 2026
Check the source to verify that this is the current v

with the number of individual defects not to exceed the quantities stated in table II.

TABLE II.—*Firing defects and limits in a lot*

Type:	*Number of defects*
Pierced case	2
Ruptured case, complete [1]	1
Ruptured case, partial [2]	2
Split case, serious [3]	1
Split case, nonserious [4]	3
Cracked rim [5]	1
Failure to chamber	2
Failure to extract	2
Misfire	2
Bullet remaining in bore	0

[1] A complete rupture is a circumferential separation whereby the case separates into two pieces.
[2] A partial rupture is a circumferential separation less than 360 degrees in magnitude.
[3] A serious split is one which is through the head or through the sidewall within ⅛ inch from the face of the head.
[4] A nonserious split is one which occurs in the body of the case.
[5] A cracked rim is a crack ⅛ inch long or less, a portion of which is under the firing pin indent.

F. METHODS OF INSPECTION AND TEST

F–1. *Inspection and tests.*—

F–1a. *Place of inspection.*—Inspection and testing may be done at the plant of the contractor or point of delivery, or both, unless otherwise specified in the contract or purchase order.

F–1b. *Lot.*—The standard size lot shall be not more than 5,000,000 nor less than 2,000,000 rounds. However, when requested by the contractor and approved by the Government inspector, the lot may contain less than 2,000,000 rounds. The size of a lot may be increased by inclusion of the cartridges of the sample which are returned to the contractor, as provided in paragraph F–1c (2). Each lot shall consist of cartridges fabricated by the same processes from like materials and identified by lot number furnished by the contractor. The test sample shall not be considered as deliveries under the contract.

F–1c. *Inspection sample.*—

F–1c (1). Inspection and testing shall not be started until each lot is completed and packed, except that packing cases need not be closed until the lot is accepted.

F–1c (2). The Government inspector shall select 10,000 rounds representative of each lot for the purpose of inspection and test. Cartridge cardboard boxes (50 rounds) shall not be opened except to obtain the cartridges needed for tests required by tables III, IV, and V. The inspection sample shall be marked by the inspector so that the unopened cartons can be identified with the lot number. That portion of the sample which has not been required for the test or retest shall be returned to the contractor, provided no cartridges have been removed from the cardboard boxes. All sample cartridges, other than those in unopened cardboard boxes, remaining after completion of the test shall be destroyed. The test sample shall not be considered as deliveries under the contract (par. F–3, National Military Establishment Spec. JAN–A–625, is not applicable)

Source: https://assist.dla.mil -- Downloaded:
Check the source to verify that this is the curre

TABLE III.—*Acceptance samples—first test.*

Test	*Number of rounds*
Bullet pull	50
Velocity [1]	24
Pressure [1]	24
Accuracy (100 rounds plus 10 warmers in each of 2 weapons)	220
Noncorrosiveness	10
Heat test:	
Accuracy (100 rounds plus 10 warmers in each of two weapons)	220
Velocity (24 rounds of normal test sample plus 24 rounds of heat test sample)	48
Function and casualty:	
Rifle, U. S., caliber .22, M2	50
Rifle, Mossberg, caliber .22, M44	50
Shotgun, rifle, Stevens, .410 gage/caliber .22, over and under barrel	50

[1] Velocities and pressures are measured simultaneously.

F–1d. *Visual inspection.*—

F–1d (1). The visual inspection sample shall consist of 3,000 rounds and shall be taken from the rounds selected in accordance with the provisions of paragraph F–1c (2). Subsamples of 300 rounds each shall be taken from the visual inspection sample and inspected for the defects listed in table I according to the sequential sampling plans of table IV.

F–1d (2). All rounds in the first subsample shall be inspected, but the inspection of second or later subsamples may be discontinued if at any time the number of defectives found be equal to or greater than the applicable rejection number. If the number of defectives found (each class, separately) be equal to or less than the applicable acceptance number, the lot shall be passed for the class of defects involved. If the number of defectives found (either class) be equal to or greater than the applicable rejection number, the lot shall be rejected. If the number of defectives found be between the applicable pair of acceptance and rejection number, the inspection of subsamples shall be continued until a decision is reached.

TABLE IV.—*Normal visual acceptance inspection.*

Subsample		Subcombined samples (major defects)			Subcombined samples (minor defects)	
Number	Size	Size	Acceptance number	Rejection number	Acceptance number	Rejection number
1st	300	300	(*)	3	1	8
2d	300	600	0	4	6	13
3d	300	900	1	5	10	18
4th	300	1200	2	6	15	22
5th	300	1500	3	7	19	27
6th	300	1800	4	7	24	31
7th	300	2100	5	8	28	36
8th	300	2400	6	9	33	41
9th	300	2700	8	9	41	42

*Acceptance not permitted until 2 subsamples have been inpsected.

Source: https://assist.dla.mil -- Download
Check the source to verify that this is the c

F–1d(3) For the purpose of visual inspection, the term "defective" shall have the meaning given in note 4, table I, National Military Establishment Specification, JAN–A–625.

F–1d(4) The Government inspector shall compute the average percentage of defective cartridges (majors and minors separately) encountered in first subsamples representing each five consecutive lots, regardless of whether the lots from which these first subsamples were taken were accepted or rejected. If the average percentage obtained be equal to or less than limiting values,

Majors—.34 percent
Minors—1.05 percent

the normal inspection procedure of table IV shall be followed in inspecting the next series of lots presented. If the average percentage for either or both classes of defects be greater than the limiting value, inspection of subsequent lots shall be made according to the applicable parts of table V. Change from the procedure of table IV to the procedure of table V shall apply for majors only, minors only, or both, depending on whether the limiting values are exceeded in the case of majors, minors, or both classes of defects, respectively.

TABLE V.—*Tightened visual acceptance inspection.*

Subsample		Subcombined samples (major defects)			Subcombined samples (minor defects)	
Number	Size	Size	Acceptance number	Rejection number	Acceptance number	Rejection number
1st	300	300	(*)	2	0	4
2d	300	600	(*)	2	1	5
3d	300	900	0	3	2	7
4th	300	1200	1	4	4	9
5th	300	1500	2	5	6	11
6th	300	1800	2	5	8	13
7th	300	2100	2	5	10	14
8th	300	2400	3	5	12	15
9th	300	2700	4	5	15	16

*Acceptance not permitted until 3 subsamples have been inspected.

F–1d (5). Transition from the procedure of table V to the procedure of table IV shall be made only when the computed average percentage or percentages of defectives found in subsamples representing at least five consecutive lots are better than acceptable quality level.

F–1d (6). If the subsamples which have been visually inspected contain insufficient rounds for the bullet pull and firing tests, the additional rounds required shall be taken from the rounds selected in accordance with the provisions of paragraph F–1c (2). Each round shall be inspected visually for the defects listed in table I. Defectives found shall not be charged against the lot but shall be removed and reported.

F–2. *Bullet pull test.*—From the first lot of cartridges ordered on any contract or purchase order, 100 cartridges shall be taken from

Source: https://assist.dla.mil -- Down
Check the source to verify that this is t

sample (par. F–1c (2)) and the bullets pulled, using standard type physical testing machine, either hydraulic or mechanically powered, which will accurately register the force required to pull the bullet from the cartridge case. The cartridges may be held either in commercial type fixtures or in the fixtures listed on drawings FA–26629 to FA–26632, inclusive. All subsequent lots shall be tested for bullet pull as above, except that at the discretion of the inspector, the number of cartridges from each lot may be reduced to ten.

F–3. *Pressure firing test.*—Twenty-four cartridges shall be fired in a pressure gage, drawing 49–15–1, with velocity taken simultaneously. The first, second, thirteenth, and fourteenth shots are discarded as they are intended to seat the pressure piston with the obturating cup. Obturating cups are changed after each 12 rounds. The use of Russian wax or Navy oil, symbol 3080, aviation, or equivalent, in the obturating cup is optional. Pressure coppers shall be uncompressed and 0.146 inch diameter by 0.400 inch length in dimensions. Pressures obtained, when corrected in accordance with provisions of paragraph F–4, shall not exceed pressure limits stated in paragraph E–3.

F–4. *Velocity firing test.*—The velocity obtained under paragraph F–3, when corrected in accordance with the following provisions, shall comply with the provisions of paragraph E–4.

F–4b. Reference ammunition loaded to an instrumental velocity at 25 feet from the muzzle of 1125 plus or minus 25 feet per second within an average pressure limit of 24,000 pounds per square inch will be supplied and assessed by the Sporting Arms and Ammunition Manufacturers' Institute. This ammunition will be used for qualifying the gages and for making corrections in velocity and pressure firings in acceptance tests. Corrections in both velocity and pressure will be applied to the ammunition undergoing test, plus or minus as the case may be, based upon the deviation of the reference ammunition fired at time of test from the Sporting Arms and Ammunition Manufacturers' Institute assessed value of the reference ammunition in use at the date of signing of the ammunition purchase contract. Weapons will be disqualified from use for velocity and pressure measurements in acceptance tests if, when tested with reference ammunition, they deviate more than plus or minus 35 feet per second or plus or minus 3,000 pounds per square inch from the assessed velocity and pressure of such ammunition. Reference ammunition will be withdrawn from use in acceptance tests and replaced immediately by the Sporting Arms and Ammunition Manufacturers' Institute when, based upon the regular, periodic, round robin interrange firings, the average results are found to vary by more than plus or minus 25 feet per second from the assessed velocity or to exceed 24,000 pounds per square inch in pressure.

F–5. *Accuracy firing test.*—Two hundred cartridges shall be fired to obtain 20 targets of ten shots each at a distance of 100 yards over an indoor range. These targets are evaluated for accuracy by finding the extreme spread, measured center to center of impact. One hundred rounds of the test ammunition shall be fired in Rifle, Winchester, Caliber .22, M52, Heavy Barrel Type, and 100 rounds of the test ammunition in Rifle, Remington, Model 37. Ten warming rounds shall be fired in each weapon immediately prior to firing for record. Each weapon shall be qualified for use at time of acceptance tests, by firing two targets of ten shots each, using check ammunition of

Source: https://assist.dla.mil -- Downloaded: 202
Check the source to verify that this is the current

known previously satisfactory accuracy performance. If the accuracy obtained with the check ammunition when so fired complies with the provisions of paragraph E-3, the weapon may be used for acceptance and results of the ammunition undergoing test when fired therein shall be considered valid. Mounting of the rifles shall be in an approved rest. The contractor's procedure for firing accuracy may be used. The inspector shall witness the firing.

F-6. *Noncorrosiveness test.*—

F-6a. Ten rounds shall be fired in a barrel, caliber .22, which, without cleaning the bore, shall then be placed in a humidity oven, with circulating air, so that the axis of the bore is parallel to the path of air.

Temperature of oven	115° to 120° F.
Relative humidity of oven	75 to 80 percent.
Time of exposure	48 hours.

F-6b. After 48 hours exposure to humid air, the barrel shall be removed and the bore visually examined for rust. Several clean cloth patches shall be passed through the bore and examined for evidence of rust as distinguished from powder residue.

F-6c. A noncorrosiveness test shall be performed for at least one ammunition lot from each cartridge type covered by a procurement contract. Provided no changes in powder or primer compositions have been made, additional noncorrosiveness tests for the same cartridge type may be waived, at the discretion of the contracting officer, after acceptance of the first ammunition lot.

F-7. *Heat test.*—

F-7a. *Heat test sample.*—Three hundred cartridges (six boxes of 50 cartridges each) shall be selected. Four cartridges shall be selected from the middle of each box and stored at 70° F. at ambient humidity as a normal sample for velocity test. These normal sample cartridges shall be replaced in the boxes by other cartridges of the same type from the same ballistic sample. The 300 cartridges shall be subjected to a temperature of 115°–120° F., with a relative humidity of 75–80 percent for a period of 48 hours, after which they shall be normalized by storing at 70° F. at ambient humidity for at least 2 hours. After normalization, 4 cartridges shall be selected from each box and the 24 cartridges shall constitute the heat test velocity sample.

F-7b. *Accuracy.*—Two hundred cartridges shall be selected at random from the heat test sample and 10 targets of 10 shots each fired at a range of 100 yards, in each of two rifles, Winchester, caliber .22, M52, Heavy Barrel Type, and Remington, caliber .22, M37, which have been qualified for use by test with check ammunition of known accuracy performance, as provided in paragraph F-8. Immediately prior to firing for record, 10 warming rounds shall be fired in each weapon.

F-7c. *Heat velocity firing test.*—The 24 cartridges set aside as a normal sample shall be fired for velocity, with pressure, in a velocity and pressure barrel, drawing 49–15–1, followed by the 24 cartridges set aside from the heat test sample. Gas checks shall be changed after each 12 cartridges. The first, second, thirteenth and fourteenth velocities and pressures from each series shall not be included for record as they are gas check seating shots. Pressures shall be recorded for information only.

Source: https://assist.dla.mil -- Downloaded
Check the source to verify that this is the curr

F–8. *Function and casualty firing test.* The cartridges in the quantities shown in table III under the heading of "No. of rds." shall be fired in the corresponding weapons listed for function and casualty. In repeating type weapons, the method of firing shall be both magazine and single cartridge feeding. The quantities to be fed by each method shall be specified by the Government inspector. In auto-loading-type weapons, the quantities to be fired shall be magazine fed. Firing the weapon from a fixed rest, shoulder or hand, is optional. The ammunition shall not be penalized for any defect caused by a defective weapon.

F–9. *Defective small arms.*—See National Military Establishment Specification JAN–A–625.

F–10. *Retests.*—See National Military Establishment Specification JAN–A–625. Double the number of rounds listed in table III of this specification shall be used in retests.

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT

G–1. *Packaging and packing.*—

G–1a. Unless otherwise specified, cartridges shall be packed as follows: 50 in a cardboard box, 10 boxes in a carton, and 20 cartons in a wooden packing box, a total of 10,000 cartridges per box unit.

G–1b. Unless otherwise specified by contract or purchase order, the wooden packing box shall be a commercial one, as furnished by the contractor. When required by the contract or purchase order, the box shall be strapped with two bands on each end portion of the box, using either wire or flat strapping.

G–2. *Marking.*—

G–2a. Each cardboard box, carton, and exterior shipping container shall be marked on the outside with the manufacturer's name or identifying mark. The ammunition lot number shall be placed on each cardboard box, and exterior shipping container. Marking of the cardboard box may be either inside or outside, at the manufacturer's option, but marking of the exterior shipping container shall be upon the outside.

G–2b. Unless otherwise specified, marking of shipments for the Army shall be in accordance with U. S. Army Specification No. 100–2; for the Navy, in accordance with the Navy Shipment Marking Handbook.

H. NOTES

H–1. The Government will furnish, free of charge, the following equipment for use in acceptance testing: all rifles, gages, shotguns, and pistols, including all mounts, tools, gages, pressure cylinders with tarage table, and accessories on the drawings listed in paragraph A–1.

H–2. Provisions should be incorporated in each inquiry or invitation for bids as follows:

"If the bidder desires to furnish a cartridge which in any way differs from the requirements of this specification, a statement to that effect shall be submitted with the bid. This statement shall describe fully the proposed difference. In addition, he shall submit full and sufficient evidence that such proposed cartridge is equal in all respects to the cartridges covered by this specification. The bidder shall submit samples of the proposed cartridge, at the request of the contracting officer, for such inspection and tests as are considered necessary to

Source: https://assist.dla.mil -- Downloaded: 2026-1
Check the source to verify that this is the current vers

determine whether the proposed ammunition complies with the requirements of the specification."

At the discretion of the contracting officer, alternate ammunition may be accepted. However, in all such cases, all bids should be withdrawn and a new proposal covering the alternate ammunition should be issued.

H–3. Ordinarily, the standard for visual defects used will be that furnished by the Sporting Arms and Ammunition Manufacturers' Institute. In the absence of standards so furnished, the contracting officer shall establish and use his own standards for the classification of defects listed in table I.

H–4. Rifles and other test weapons shall be checked for proper dimensions and functioning before being issued for test purposes. The items to be checked include:

(1) Firing pin protrusion.
Rifle, U. S., M2—0.032 to 0.036 inch.
(2) Head space—0.044 to 0.046 inch.
(3) Correct shape of firing pin.
(4) Correct chamber and bore dimensions.
(5) Proper function of extractor.
(6) Proper function of magazine.

Such test weapons will not be weapons of exceptional quality but are to be standard stock.

H–5. Procedure and technique of ballistic testing will be according to ORD–M608–PM, Vol. III, where applicable to rim fire, caliber .22 ammunition.

H–6. Copies of specifications and drawings may be obtained from the Chief of Ordnance, Department of the Army, Washington 25, D. C.

H–7. Copies of the Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Department of the Navy, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Center, Norfolk 11, Va.

H–8. Copies of this specification (required by Naval activities) may be obtained upon application to the Supply Officer in Command, Naval Supply Center, Norfolk 11, Va. Both the title and identifying symbol number should be stipulated when requesting copies.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

ARMY: O

NAVY: OS

[JAN–C–716]

Source: https://assist.dla.mil -- Downloaded: 20
Check the source to verify that this is the current
555270